PHYSICAL CHEMISTRY CUMULATIVE EXAM
November 17, 2000

(1) Estimate $\Delta S^{o\#}$ for the reaction

$$A + B \overset{K}{\rightleftharpoons} (AB)^{\#}$$

where A is an atom, B is a diatomic molecule and $(AB)^{\#}$ is a nonlinear triatomic molecule. You may assume that S^{o}_{vib} (one degree of freedom) = 1 eu, S^{o}_{rot}(one degree of freedom) = 30 eu and S^{o}_{trans} (one degree of freedom) = 50 eu. Discuss your answer with respect to sign and numerical value.

(2) The reaction between O_3 and CS_2 was studied at 29 °C with excess CS_2. From the following data determine the order of the reaction and calculate the rate constant.

t, min	0.0	0.5	1.0	2.0	3.0	4.0
P_{O_3}, torr	1.76	1.04	0.79	0.52	0.37	0.29

(3) Determine a, b and c for the rate equation

$$\text{rate} = kC^{a}_{A}C^{b}_{B}C^{c}_{C}$$

from the following data:

rate × 10^5, $M\ s^{-1}$	5.0	5.0	2.5	14.1
$C_{A,0}$, M	0.010	0.010	0.010	0.020
$C_{B,0}$, M	0.005	0.005	0.010	0.005
$C_{C,0}$, M	0.010	0.015	0.010	0.010